MOOCs Data Analytics and Machine Learning Project Report

Data Analytics and Optimization using Python

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Project Objectives

The main objectives of this project are:

Perform complete descriptive analysis, Handle missing values and duplicate records, Apply preprocessing techniques to prepare the data, Visualize important patterns and relationships inside the dataset, Compare the performance of five machine learning classification models, Apply clustering techniques using KMeans, Determine the most important features affecting predictions.

Dataset Description

The dataset includes several features related to:

- Student sentiment, Confusion level
- Urgency level, Number of reads
- Up-votes, Course type
- Questions and opinions

The target variable used in this project is: ***Opinion(1/0)***

This target converts the problem into a classification problem.

Libraries Used

The following Python libraries were used during the project:

- Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

These libraries were used for:

- Data preprocessing, Visualization, Machine learning, Clustering, Model evaluation

Data Preprocessing

Handling Missing Values

Two imputation techniques were used:

- Median imputation for numerical columns.
- Most frequent value (mode) imputation for categorical columns.

Removing Duplicate Records

Duplicate rows were detected and removed using:

df.drop_duplicates(inplace=True)

Encoding Categorical Variables

Machine learning models cannot directly process text values. Therefore, categorical columns were converted into numerical values using Label Encoding.

Feature Scaling

StandardScaler was used to normalize feature values before training the models.

Feature scaling is especially important for algorithms such as:

- Support Vector Machine (SVM)
- Logistic Regression
- KMeans Clustering

Exploratory Data Analysis (EDA)

Course Type Distribution, Sentiment Distribution

Urgency Distribution, Confusion Distribution

Boxplots, Correlation Heatmap, Scatter Plot

Machine Learning Models

Five machine learning classification models were implemented and compared.

The models are:

Naive Bayes, Support Vector Machine (SVM), Random Forest, AdaBoost, Logistic Regression

The dataset was divided into:

- 80% training data
- 20% testing data

using train_test_split().

Model Evaluation

Naive Bayes

Support Vector Machine (SVM)

Random Forest

Ada Forest

AdaBoost

Logistic Regression

Model Comparison

Confusion matrix

A confusion matrix was generated for the Random Forest model.

The confusion matrix helps evaluate:

- Correct predictions, Incorrect predictions, Classification performance for each class

Clustering Analysis

Unsupervised learning was also applied using KMeans clustering.

Clustering helps discover hidden groups and patterns inside the dataset without using target labels.

KMeans Clustering

KMeans clustering was applied using:

KMeans(n_clusters=3)

The algorithm grouped similar records into clusters.

Elbow Method

The Elbow Method was used to determine the optimal number of clusters.

The inertia values were plotted for different cluster counts.

Cluster Visualization

Clusters were visualized using a scatter plot between:

Sentiment,& Confusion

Different colors represented different cluster

Feature Importance

Feature importance analysis was performed using the Random Forest model.

This analysis identifies which features contribute most to prediction accuracy.

The top important features were visualized using a bar chart

Results and Discussion

The project demonstrated that preprocessing and visualization significantly improve machine learning performance.

The clustering analysis]revealed meaningful patterns in student behavior and interaction.

The project successfully combined:

- Descriptive analytics, Machine learning classification
- Clustering analysis, Visualization techniques

Conclusion

The project included:

- Data preprocessing, Missing value handling
- Duplicate removal, Data visualization
- Classification models, Clustering techniques
- Model comparison, Feature importance analysis

The final results demonstrated the importance of preprocessing and model evaluation in achieving accurate predictions.

The project also highlighted how machine learning can help analyze student behavior and interaction patterns in online learning environments.

Full Project Code

Paste the complete Python project code here.

Paste your full code here